

Solution of shortest path genetic algorithm for regional traffic network based on Petri Net

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Abstract

With the development of China's social economy and science and technology, the scale of urban traffic network has become bigger and bigger. How to quickly find the shortest path in the regional traffic network with dense roads and complex traffic nodes has become an urgent concern for current traffic travelers. In order to overcome the shortcomings of traditional shortest path algorithms such as Dijkstra in its long solution time and low accuracy when solving large-scale urban traffic networks, based on the constructed traffic network topology, this paper proposes a shortest path model based on Petri net and traffic network weighted S-graph, taking the characteristics of urban traffic network into consideration, this paper establishes an improved shortest path Genetic Algorithm (GA) based on the traditional GA. The research results show that the improved GA has the advantages of high speed and accuracy when solving the shortest path of the regional traffic network, and can quickly find the shortest path with smaller error compared with the actual value. Once again, it's proved that constructing an efficient and accurate shortest path search algorithm is very important. The research results can provide a theoretical basis for the shortest path search of regional traffic network.

Keywords – petri net, shortest path, weighted s-graph, improved genetic algorithm (GA)

1. Introduction

With the continuous development of the national economic level, large population gather in cities, the population density in urban areas has increased dramatically [1-2]. This has led to the continuous expansion of urban roads and regional traffic networks. According to reports, the number of vehicles in China had exceeded 330 million by the end of 2018 [3]. The continuous growth in the number of vehicles has led to a growing demand for travel capacity of urban traffic networks. Alleviating urban traffic conditions can usually start with two aspects: accelerating the construction of urban road infrastructure, and strengthening the management of urban traffic [4]. Wherein, accelerating the expansion of urban roads is the most direct way to solve traffic congestion. However, within cities, due to the limited land resources, it is often not easy to build large-scale urban roads, which brings great pressure to the urban traffic network facilities under existing conditions [5].

Therefore, at present, as the contradiction between traffic resources and transportation demand is becoming increasingly prominent and the traffic congestion becomes worse, it is urgent for us to combine advanced science and technology to alleviate urban traffic congestion through reasonable traffic guidance, thus facilitating the work and life of the traveler and building cities suitable for people to live in [6].

Intelligent transportation is a product of adapting to the development of the times, it integrates computer processing technology into the traffic management system, and establishes an intelligent traffic network which can update road conditions in real time [7]. Through GPS and other technologies, it can provide real-time road condition information for travelers, and give optimal path instructions, which could help travelers clearly know which path to take so as to better avoid traffic congestion and reach the destination safely and quickly [8]. The search of shortest path is the core content of the intelligent transportation system. The efficiency and correctness of the solution are of great practical significance for effectively alleviating traffic problems. At present, the commonly used shortest path search algorithms show some defects and shortcomings when solving large-scale urban traffic nodes, so it is urgent to combine the characteristics of regional traffic networks to find a new optimal path algorithm for urban traffic networks [9].

In summary, the research on the optimal path of urban regional traffic network has very important theoretical and practical significance. Improving the deficiencies of existing algorithms can provide theoretical guidance for the advanced urban road traffic management systems, and it helps to alleviate regional traffic congestion and improve travel efficiency. Therefore, based on the introduction of the Petri net and traffic network weighted S-graph theory model, the urban traffic network is modeled in this paper, and the traditional GA is improved based on the characteristics of urban traffic network. Moreover, this study establishes the shortest path search algorithm based on improved GA and uses the example of a regional urban traffic network to verify the model and the algorithm. The research results show that the improved GA can accurately and quickly search for the shortest path of urban traffic network.

2. Urban traffic network modeling based on Petri Net

In order to solve the shortest path problem of the huge-scale urban traffic network, it is necessary to combine the characteristics of dense roads in cities to design an algorithm that can calculate the shortest path problem quickly, thereby, this paper proposes the S-graph shortest path algorithm based on Petri net.

2.1. Overview of urban traffic network

The intersections in the city usually have various forms, such as X-shape, Y-shape, cross-shape, and T-shape, etc. [10]. Figure 1 shows an urban regional traffic network with 12 intersections, wherein the number in the circle is the number of the intersection.

2.2. Description of Petri Net

Petri net can be used to describe distributed systems, it usually includes two types of elements, the element library place $S(place)$ and the element transition $T(Transition)$, which are used to simulate the structure and running of the system [11-12], wherein S is the state and T is the state change.

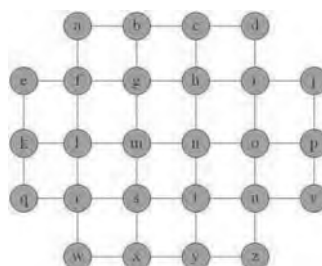


Fig. 1 - An urban traffic network map with 12 cross-shape intersections

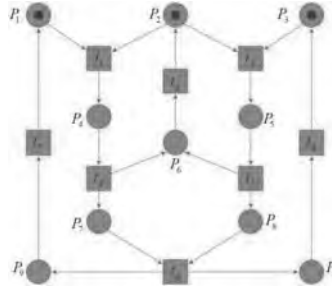


Fig. 2 - Schematic diagram of the Petri network

Usually, $\square$ represents the node of the transition, and $\circ$ represents the node of the condition. A petri net is shown in Figure 2.

Definition 1: $N(S, T, F)$ is used to represent a ternary network. When the network satisfies the conditions in Formula (1), it constitutes a Petri network.

$$\begin{cases} S \times T \neq \emptyset \\ S \cap T = \emptyset \\ F \subseteq (P \times T) \cup (T \times P) \\ \text{dom}(F) \cup \text{cod}(F) = P \cup T \end{cases} \quad (1)$$

where, F represents a directed arc between (x, y) , $\text{dom}(F) = \{x \in S \cup T | \exists y \in S \cup T: (x, y) \in F\}$; $\text{cod}(F) = \{x \in S \cup T | \exists y \in S \cup T: (y, x) \in F\}$.

2.3. Traffic network weighted S-graph theory model

Definition 2: $N=(S, T, F, K, W, M)$ represents a hexahydroxy network. When the network satisfies the conditions of Formulas (2), (3), and (4), it constitutes a S-graph model [13].

$$\begin{cases} W : T \rightarrow \{1, 2, 3, \dots\} \\ K : S \rightarrow 1 \\ M : S \rightarrow \{0, 1\} \forall s \in S : M(s) \leq K(s) \end{cases} \quad (2)$$

where, W is a weight function and K is a capacity function.

For the transition, $t \in T$:

$$\begin{cases} \forall s \in t : M(s) \geq 1 \\ \forall s \in t - t : M(s) = 1 \\ \forall s \in t \cap t : M(s) \leq K(s) \end{cases} \quad (3)$$

If t happens from M and obtains M' , then for $\forall s \in S$, there are:

$$M'(s) = \begin{cases} M(s) - 1 & \text{if } s \in t - t \\ M(s) - 1 & \text{if } s \in t - t \\ M(s) - 1 & \text{if } s \in t \cap t \\ 0 & \text{else} \end{cases} \quad (4)$$

3. Shortest path search based on GA

3.1. The basic idea of GA

GA draws on the idea of natural inheritance, through certain coding methods, it transforms the variables of a problem into an initial population which is composed of multiple individuals.

According to the natural selection principle, in the nature, the fittest survives and the un-fitted eliminated, GA continuously selects the population with high adaptability, then through crossover and mutation rules, it gradually evolves into the solution set of the optimal solutions [14-15].

In general, the offspring population generated by the selection, cross, and mutation processes are good individuals of parent population, it can both inherit good qualities of the parent and evolve continuously, the expression of GA is shown as Formula (5).

$$GA = (C, F, P(0), PopSize, Px, Pm, T) \quad (5)$$

where, $C(coding)$ represents the coding mode, such as binary coding; $F(fitness)$ represents the fitness function, $P(0)$ represents the initial parent population, $PopSize$ represents the size of the population, Px and Pm represent the crossover probability and mutation probability, respectively, T represents the termination condition of the algorithm, which is generally represented by the number of iterations.

3.2. Implementation of shortest path search GA

The urban traffic network has the following characteristics [16]:

- (1) The number of nodes in the traffic network is large, and the basic GA requires a large running storage space;
- (2) The number of nodes in the path may change, and unable to adapt to the characteristics of the length of each individual in the binary code;
- (3) The random sequence of the path of the traffic network is inconsistent, so the traditional binary coding may produce non-feasible solution;
- (4) The starting point and ending point of the path of the traffic network need to be changed according to the change of the path search requirement, and the initial population of traditional basic GA is randomly generated.

Therefore, based on the traditional GA, the improved shortest path search GA is shown in Figure 3.

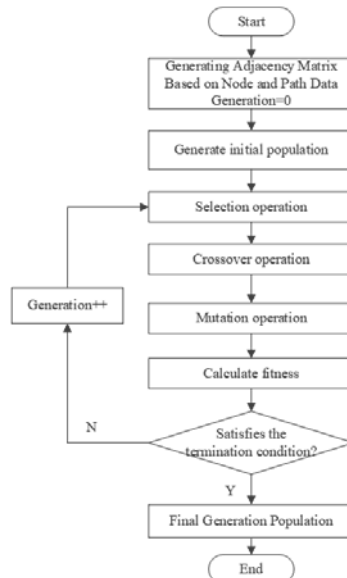


Fig. 3 - Implementation process of shortest path search GA

The improved GA is mainly to overcome the following shortcomings of the basic GA [17]:

- (1) Selection process: it is very easy to lose the optimal individual, resulting in the algorithm falling into local optimum;
- (2) Crossover process: the method is relatively simple, the diversity of the population is not rich enough, and the integrity of the path is easily destroyed;
- (3) Mutation process: the modification of the genes is too much, and it is easy to destroy individuals with good fitness.

Therefore, for the above three deficiencies, the following improvements are made:

- (1) Selection process

On the basis of roulette selection, the elite strategy is added to ensure that the optimal individual in each generation will not be lost.

- (2) Crossover process

Assume two parents are p_1, p_2 , two intersection positions a_1, a_2 are generated randomly, if the value of p_1 at position a_1 is x , then in p_2 , it needs to traverse from position a_2 to find a gene position with a value of x , after the position is found, crossover with p_1 from the position.

- (3) Mutation process

The traditional mutation method begins with a randomly generated mutation position m_x and regenerates a path to the ending point. This equation is extremely easy to destroy excellent gene individuals. Therefore, in order to avoid destroying the original excellent gene individuals, an improved two-point mutation method is adopted to regenerate the path between the two points.

4. Example analysis

4.1. Example description

A regional traffic network of a city has 251 nodes and 362 road sections. The topology is shown in Figure 4.

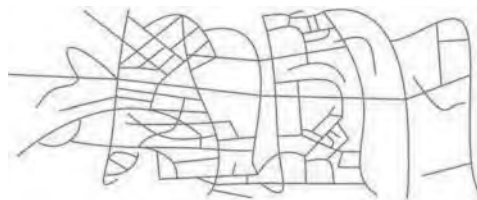


Fig. 4 - XX city urban traffic network topology

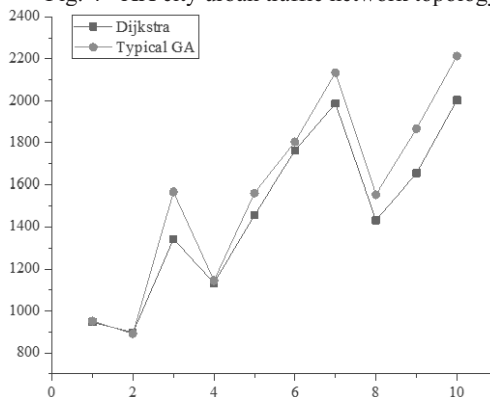


Fig. 5 - Point plot of the search length of the two algorithms

Tab. 1 - Comparison of the results of the two algorithms

Path Start ID	1	52	93	76	58	82	104	124	152	164
Path End Path	47	79	125	149	186	224	251	187	203	251
Dijkstra	947	897	1342	1134	1456	1763	1987	1432	1655	2003
Typical GA	952	892	1567	1145	1561	1804	2134	1554	1867	2214

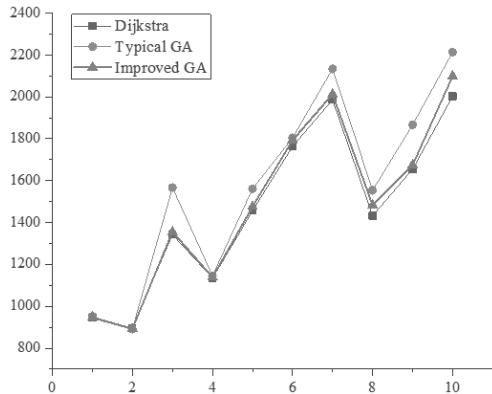


Fig. 6 - Point plot of the search length of the three algorithms

4.2. Example results and discussion

Two sets of paths were randomly selected, and the solutions of the traditional Dijkstra algorithm [18] and the improved GA constructed in this paper were compared, the solution results are shown in Table 1. Figure 5 visually depicts the differences in search length between the two algorithms.

The results in Table 1 and Figure 5 show that when the urban traffic network contains more nodes, the path obtained by the basic GA is not the actual shortest path; when the distance between the two points is shorter, the results of basic GA and the Dijkstra algorithm are almost coincident.

Therefore, the improved GA was further applied and the improved search results were obtained as shown in Figure 6.

Tab. 2 - Comparison of shortest path search results (time and length) of GAs before and after improvement

Path	Typical GA		Improved GA		Actual length (m)
	Length (m)	Solving time (s)	Length (m)	Solving time (s)	
1-49	972	0.424	972	0.43	972
2-62	924	0.29	924	0.288	924
7-48	773	0.607	773	0.587	773
39-121	1282	0.53	1282	0.54	1282
63-99	1340	0.5	1338	0.5	1338
82-198	1246	0.665	1230	0.671	1230
73-168	1805	0.836	1805	0.836	1751
7-147	1974	0.74	1959	0.708	1959
39-156	1225	0.671	1217	0.671	1217
1-192	3082	0.711	3082	0.79	2953
65-173	2516	0.595	1477	0.431	1444
12-188	3708	0.699	2264	0.535	2264
143-232	2703	697.05	2163	0.781	2163
39-175	2394	0.576	1858	0.654	1858
173-251	2237	0.623	1779	0.462	1779

It can be clearly seen from Figure 6 that the improved GA is superior to the traditional GA in searching for the shortest path. Further, several starting points and ending points were randomly selected, and the GAs before and after the improvement were compared in terms of the search time and the shortest path. The results are shown in Table 2.

It can be further proved from Table 2 that the GAs before and after the improvement have little difference in the solution time, but the improved GA is superior to the basic GA in the shortest search path length obtained, which is almost consistent with the actual length.

5. Conclusion

Considering the importance of shortest path search in urban traffic intelligent network system, this paper focused on the solution of regional traffic network and drew the following conclusions:

(1) Based on the dense urban traffic network, in order to construct an algorithm for solving the shortest path search problem efficiently, this paper proposed a shortest path model based on the Petri net and the traffic network weighted S-graph.

(2) Based on the shortcomings of traditional GA in the shortest path search of urban traffic network, an improved shortest path GA was established in the paper.

(3) Using the traffic network topology data of a certain city, this paper compared the Dijkstra algorithm, the basic GA and the improved GA from two aspects of solution time and the shortest path, and proved the superiority of the improved GA.

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